A SEMINAR REPORT

ON

"ANALYSIS AND OPTIMIZATION RESEARCH OF HIGH-PRESSURE CRYOGENIC LIQUID ROCKET ENGINE"

B.TECH- IV (MECHANICAL)

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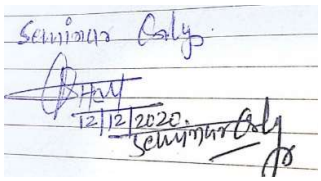

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ABSTRACT

Due to their high specific impulse property, Hydrogen-Oxygen systems are often used to increase launcher performance for cryogenic engines. They will probably continue to be the preferred option for the next decades. To get insight into complex processes involved in such propellants' combustion, research efforts have been conducted in both theoretical and experimental activities.

First, general design issues of liquid propellant engines are addressed, demonstrating a clear need for a new approach in engine development, taking into account the complex chemical and physical processes in rocket engines considering the following problems, namely Combustion chamber heat load, Combustion instability, and Performance Transients. A brief description of the organization of the research programs is followed by a short article on the test facilities and the experimental setup in use. Examples of experimental and numerical achievements applying sophisticated optical diagnostic tools such as Cold flow injection and atomization, Particle Tracking Velocimetry, and Turbulent combustion diagnostics are discussed. Thermomechanical, Design, Material Analysis, and Optimization techniques are discussed, and finally, the paper summarises the results achieved so far and addresses the questions still open.

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ABBREVIATIONS AND ACRONYMS

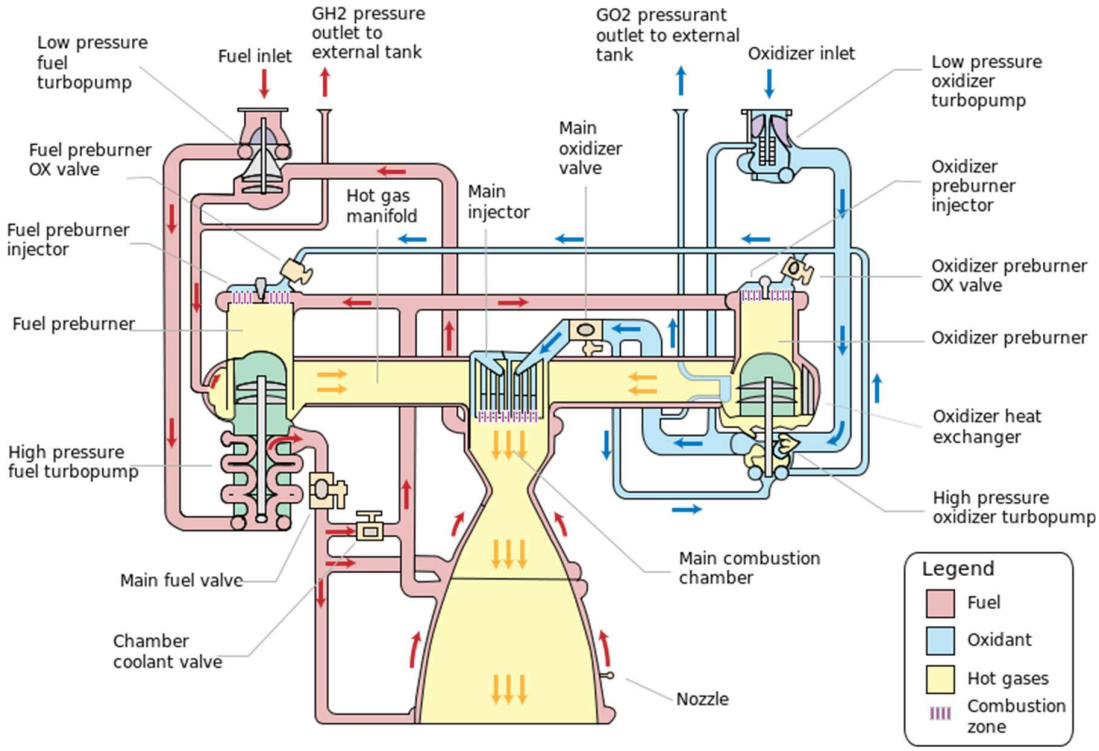
- Office National d' Etudes et de Recherches Aerospatiales (ONERA)
- Computational Fluid Dynamics (CFD)
- The German Aerospace Center(DLR)
- Particle Tracking Velocimetry(PTV)
- Liquid Oxygen(LOX)
- Memorandum of Understanding (MoU)
- National Centre for Space Studies(CNES)
- Phase Doppler Particle Analyzer (PDPA)
- Centre national de la recherche scientifique (CNRS)
- Particle Tracking Velocimetry(PTV)
- Alternate Direction Implicit (ADI)
- Liquid-propellant Rocket Engine(LRE)

Section 1: Introduction

Cryogenic technology is the process of involvement or including of usage of rocket propellants at a cryogenic temperature. It can be the combination of liquid fuels such as Liquid Oxygen (LOX) and liquid hydrogen as an oxidizer and fuel in the different mixtures or proportions. The mixture of fuels offers the highest energy efficiency for the rocket engines that produces a very high amount of thrust. Here, the oxygen remains liquid only at the temperature below (–183 C) and hydrogen at below (–253 C). This is a type of rocket engine that is functionally designed to use the oxidizer, which must be refrigerated in the liquid state. Sometimes, the liquid nitrogen is sometimes used as a fuel because the exhaust is also nitrogen. Liquid oxygen is injected below critical temperature but above the critical pressure. In our atmosphere, nitrogen is nearly about 78%. Nitrogen is a non-pollutant gas, and during exhaust, no other harmful gases are produced. Hence its efficiency is very high than any other Jet engines[1].

1.1 CONSTRUCTION AND WORKING

The figure below shows a Raptor Cryogenic Liquid-propellant Rocket Engine consisting of a staged combustion cycle for Increased engine efficiency. The main parts can be listed as



- Nozzle: The pressure generated in the combustion chamber can be used increased thrust by the acceleration of combustion gas to high supersonic velocity

- Combustion Chamber: Here, the cryogenic fuel and oxidizer are burned, and combustion takes place. The highest temperature is seen in this part of the engine

- Injector: The injector implementation in a cryogenic rocket engine determines the percentage of the theoretical performance of the nozzle that can be achieved

- Turbopumps: They receive liquid propellants at low pressure from vehicle tanks, which are then supplied to the combustion chamber.

- Preburner: the unit allows rockets to operate at a higher pressure and achieve greater levels of thrust.

- Gas Generator: The main function of a gas generator is delivery of a sufficient amount of driver gas at a designed temperature to the combustion chamber. It is governed by pressure, which generates a continuous propellant supply of thrust chamber.

Cryogenic Engines are designed for liquid fuels Hydrogen and Oxygen, which need to be held below 20°K (-423°F) and 90°K (- 297°F) to remain in a liquid state. The engine components are then cooled so the fuel does not vaporize in the lines that feed the engine. The thrust is obtained by the rapid expansion from liquid to gas, with the gas emerging from the motor at a very high speed. Cryogenic engines are the highest performing rocket engines. A disadvantage with them is that the fuel tanks are very bulky and require heavy insulation to store the propellant and cool them. The energy needed to heat the fuels is obtained from burning them once they become gas. Although their high fuel efficiency outweighs this disadvantage. The Main Space Shuttle's main engines used for lift-off are the cryogenic engines. The Shuttle's smaller thrusters for orbital maneuvering use non-cryogenic hypergolic fuels, which are compact and are stored at warm temperatures. Currently, only France, Russia, China, United States, Japan, and India have mastered cryogenic rocket technology. The cryogenic engine obtains its name from the state temperature for the storage of nitrogen. The Air moving around the engine is used to heat liquid nitrogen and to change its phase. Oxygen hydrogen fuel-oxidizer pair is very successfully used and is most prominent in the industry. Ariane 5 uses oxygen as an oxidizer and hydrogen as fuel, both stored at a cryogenic temperature in a liquid state, to produce power. The liquid nitrogen, stored at extremely low temperatures, is vaporized by the heat exchanger. Nitrogen gas formed then expands to about 700 times the volume in the heat exchanger in its liquid form. This high pressurized gas is further fed to the expander, where the pressure force is converted into Mechanical work. The Cryogenic engines have the highest thrust to efficiency ratio among every other rocket engines.

There are many reasons for which liquid-propellant rocket engines have allowed the conquest of space during the last decades. Advantages of liquid systems are their high performance as compared to any conventional chemical structure and the fact that they are highly controllable in terms of thrust modulation. Disadvantages

include the complexity and development cost of the engine. Great efforts in terms of technology investments are necessary to reach a functional maturity for systematic use. The most important work is optical diagnostics, computing power, and numerical methods, which are still new. Therefore earlier developments were mainly based on experimental data, trial and error approach. Since then, quite progress has been made in diagnostics, experimental means, computing power, and CFD methodologies. Experimental means, computing power, and CFD methods. Consequently, the design of propulsion systems has evolved from rudimentary science to a more sophisticated art. Nowadays, in the context of budget cuts and hard competition, the critical issue is how to build efficient, reliable propulsion systems at a lower cost and in a short time. To avoid these issues, a more coherent approach, including system studies, technology evaluation, and research studies, has to be involved. This paper deals with some of the recent progress achieved at DLR and ONERA on Oxygen/Hydrogen (O2/H2) combustion. Indeed, due to its property of high specific impulse, the O2/H2 system is often used to increase the launcher performance. This system will probably continue to be the preferred one for space propulsion for the next decade.

Now, before going into details on research activities on O2/H2, a brief summary is provided on combustion design issues and related physical processes[2].

Section 2. Design issues in combustion devices

Every Combustion device has different operational risks [3] specific to the application requirements. However, all of them have potential development problems such as combustion chamber heat load, startup combustion instability, low-performance shutdown, transients, and low performance.

2.1. Combustion instability

Acoustic Combustion instability has been one of the most complex phenomena in liquid rocket engines, and therefore difficult to understand completely, control, and predict particularity in the design of large scale rockets[6]. Combustion instabilities have long been recognized as a critical issue in engine development. They result from the coupling between the combustion and the fluid dynamics of the system. Several types of instability have been observed. All of them are characterized by chamber pressure oscillations, although the frequency and amplitude of these oscillations and their external manifestations normally vary with the type of instability.

Combustion instabilities have been classified into two major categories: high-frequency instability (screaming) and low-frequency instability (chugging). High-frequency instabilities result from a coupling between the combustion process and the chamber acoustics. They are the most destructive kind of instability. Despite considerable efforts spent on this subject, combustion instability mechanisms are not yet well explained. Methods for preventing combustion instability still rely on testing and the use of stabilizing devices (baffles and/or acoustic cavities). Potential mechanisms of liquid-propellant combustion instability involve a coupling between

chamber acoustics and one or more of the following processes: injection, atomization, droplet heating, Vaporization, mixing, and chemical kinetics. Nearly all of these processes have been studied experimentally or analytically as potential driving mechanisms for combustion instability. [4,5] Due to the lack of reliable predictive tools, the industry continues to use semi-empirical correlations to investigate the combustion stability of a liquid rocket engine. Experimental validation of the engine stability requirements is still needed. This leads to increased development costs and long delays. Moreover, even for an oxygen/hydrogen combustion device that is less sensitive to combustion instability, compared to other propellant combinations, baffles and/or acoustic cavities are still used to increase the stability margin. This results in more complex technology, increased weight and increased cooling requirements. LOX/H2 systems are less sensitive to combustion instability compared to storable propellant and LOX/Hydrocarbon systems. But, even if these systems have not experienced severe instability problems, their stability is not characterized by a huge range of operating conditions. Therefore, fundamental research on elementary processes occurring in a LOX/H2 combustion chamber is necessary in order to get insight into the combustion instability phenomenon and to develop more predictive analysis tools.

2.2. Combustion chamber heat load

Due to high heat transfer rates and high combustion temperature from the hot gases to the chamber wall, chamber cooling is a major issue for long-duration applications. These heat margins increase with the increasing chamber pressure needed for increasing performance, the latter being always a goal. In the case of high thrust engines, to gain more thrust temperature can reach extreme heights nearly enough to deform and make cracks in the chamber. Several techniques can be used, including transpiration cooling, film cooling, dump cooling, regenerative cooling, ablative cooling, and radiation cooling. Design of a regeneratively cooled chamber involves consideration of gas and coolant side heat transfer and wall structure requirements. The designer has to deal with many and sometimes contradictory requirements, such as long life, lightweight, low cost, and high performance and reliability. For instance, for a given material, wall design considerations would suggest reducing the chamber wall thickness for the reduction of the temperature difference between the inner coolant wall and the outer hot-gas wall for a given heat flux. However, wall thickness is determined by structural requirements to accommodate pressure and thermal stresses as well as by manufacturing feasibility limits.

Heat transfer analysis of regeneratively cooled thrust chambers is one of the most important and challenging design tasks. Wall temperature and heat flux predictions must take into account the hot-gas side heat transfer to the wall, heat conduction through the wall, and the transfer into the coolant. With operating pressure, the heat flux increases and reaches very high values at the throat, making the prediction of chamber life very sensitive to the accuracy of wall temperature evaluation [7]. If heat conduction in the wall and coolant-side heat transfer can be predicted with sufficient confidence using experimental data and/or simplified tools (1-D models), gas-side heat prediction is much more difficult because of the extremely complex flow in the chamber (high Reynolds number and thin boundary layer). The near injector zone heat transfer analysis is even more complex due to 3-D effects and interaction between the flame and the wall. Improvements of the

chamber heat load require research on both the theoretical side, with improved physical Modeling (spray combustion, turbulence, boundary layer treatment), and the experimental side, with specific tests for model validation.

2.3. Thrust chamber performance

The importance of thrust chamber performance is extremely salient because every second gained in specific impulse is mass gained in the payload. Thrust chamber performance is a function of the propellant combination and combustion efficiency, which itself depends strongly on the injector design, and nozzle performance. High levels of combustion efficiency require uniform distribution in the desired mixture ratio, fine atomization, and good mixing of propellants. Nozzle extension design has a large impact on performance for nozzles operating at sea levels. The state of the engine performance prediction is reported in [8]. The method of performance prediction starts with the calculation of ideal performance (maximum theoretical performance) and then the evaluation of all losses to be subtracted from the theoretical performance. Different types of losses due to deficiencies in energy-release caused by incomplete Vaporization and mixing and finite rate chemistry, as well as losses due to friction and divergence, can be addressed. Predicting the energy heat-release-related losses is a very difficult task because of the very complex processes that occur in the combustion chamber and their interaction: atomization, evaporation, turbulent combustion, etc. For instance, the prediction of incomplete vaporization losses needs a reliable estimate of initial droplet size distribution, along with realistic vaporization models. These phenomena are well understood and controlled, not only for a reliable assessment of heat transfer analysis and performance but also to help the designer to assess the effect of any change in either engine design (injector dimensions and spacing, for instance) or operating conditions (such as pressure or mixture ratio) on engine performance and heat loads (for a combustion chamber) and temperature stratification (for a gas generator or preburner).

2.4. Transients

Transient phases like startup and cut-off are crucial to the reliability and usage life of a rocket engine. To avoid transient mode failure, the transient behavior of each subsystem has to be controlled and predicted. Temporal evolution of physical parameters such as temperature, pressure, and mass flow rate, and ignition pressure peaks have to be specified at the design level. Risks of unsafe transients depend on the propellant type and the engine cycle. Storable propellants have the easiest transients as compared to LOX/H2 propellant systems because of the thermal chilling down constraints. As much as the engine cycle is concerned, pressure fed systems are the easiest to start. Among the pump-fed systems, the gas generator cycle has the simplest transients. The staged combustion and the expander cycle are the most complexes to create. For the LOX/Hydrogen system, chilling down, startup, and ignition are very complex. Analysis of chilling down involves Modeling of hydrogen flow in the cooling channels of the regenerative circuit. A reliable predictive tool has to deal with unsteady two-phase flow and transition from the sub- to a supercritical hydrogen flow regime. Another issue in the analysis of such flows is

the Modeling of heat transfer coefficients in two-phase flow conditions with transcritical hydrogen [9].

Analysis of startup and ignition is necessary to optimize startup sequences, avoid hard or delayed ignition, control mixture ratio evolution for devices without cooling (gas generator), etc. The flows in the regenerative circuit and the combustion chamber must be dealt with. Modeling includes unsteady two-phase flows and multi-species and chemically reacting flows. One important parameter in the startup transient is the ignition delay. Determination of this parameter is still beyond reliable prediction and is still based on experience and experimental data. Prediction of ignition involves detailed Modeling of O2/H2 chemistry at low pressure and low temperature and must also account for the igniter combustion products.

Section 3. The need for a new approach to engine development

For fulfillment of performance, stability, and engine life for an engine design, operational risks need to be avoided described in the previous sections and have, therefore, to account for all the complex chemical and physical processes which occur during the operation of such combusting devices. Moreover, in budget cuts and competition, the design engineers have to develop a reliable, high-performance engine in a small amount of time and at a low cost. A new approach is needed combining technological efforts, research investment, and cooperation between industry and research organizations. New logic in combustion device development is needed in which CFD has to be more and more involved. Indeed, delays may be reduced by reducing the total number of qualifying tests to only the identified critical operational range of the combustion device and generally, by reducing development risks by improving the capabilities of predicting tools. Validated tools developed through research efforts should not only help to make the best choices in terms of design and technology and allow prediction of operational behavior of the combustion device, but they will also be helpful in directing development tests, analyzing test results, and reducing both qualification and data analysis times, especially when a problem is encountered during engine development.

Section 4. High-pressure cryogenic combustion – approach and means

4.1. Organization

In the framework of high-pressure cryogenic propellant rocket engines for current and future launchers, technological and research activities are being conducted in both France and Germany. Both research programs involve agencies, industry, research laboratories, and universities. In France, a Research Group involving four partners: Snecma Moteurs-Rocket Engine Division, CNES, CNRS, and ONERA, has been conducting research activities on high-pressure cryogenic combustion using coaxial injection. In Germany, the national technology program "TEKAN" links basic

research into combustion modeling from universities of Aachen, Heidelberg, and Stuttgart and DLR's own research and technology program in the field with Astrium's more product-oriented activities. A close relationship exists between the two programs through a Memorandum of Understanding (MoU) on research on high-pressure cryogenic combustion between France and Germany. In practice, research results and progress are systematically acquired together or exchanged during colloquia and symposia. In practice, calibrated physical models provided by research will be implemented into CFD codes, which in turn will be implemented in the design and optimization loop of combustion devices. This ultimate objective of having validated tools at the development level requires efforts in three main areas: theoretical activities to provide models more suited to describe processes involved in high-pressure cryogenic combustion, experimental activities for validation of such models and also for code validation, and numerical analysis including code development and model implementation.

4.2. Physical processes involved in high-pressure coaxial combustion

The main elementary processes involved in a single-element shear coaxial injector can be seen in Fig. 1. High-speed hydrogen flow atomizes the low-speed liquid-oxygen jet. Ligaments and droplets formed from this primary atomization undergo secondary atomization and evaporation. Oxidizer vapor and fuel undergo turbulent mixing and combustion. A flame is anchored somewhere at the injector exit. Due to high pressure and temperature encountered in such conditions, only non-intrusive measurement methods may be used (optical diagnostics methods), as shown in Fig. 1.

For a given geometry, chamber pressure, and propellant mass flow rates, the following questions have is to be addressed:
what kind of models have to be developed to describe these elementary processes such as atomization, evaporation, mixing, and combustion under realistic operating conditions of high-pressure liquid rocket engines?

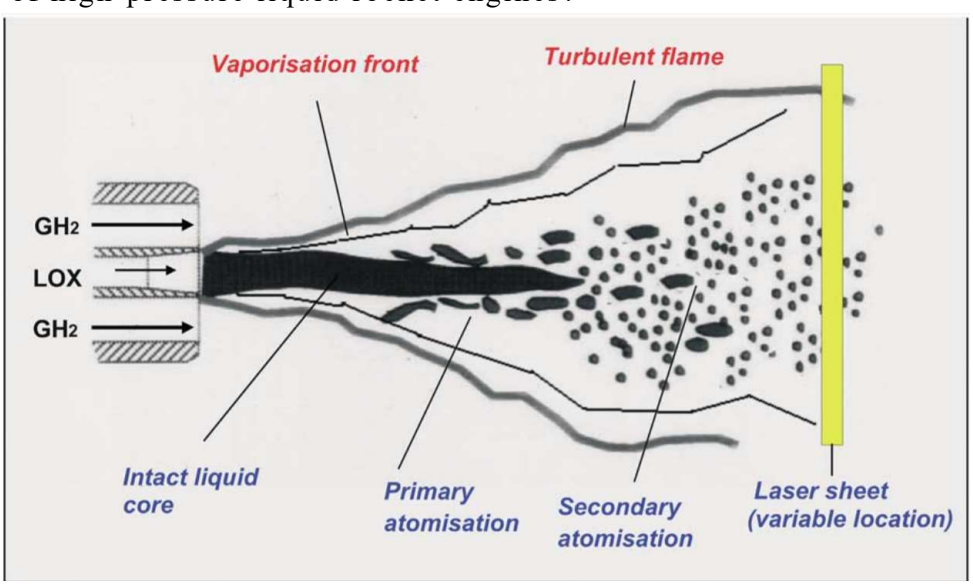


Figure 1 Elementary processes involved in a high-pressure coaxial injection

What are the governing parameters that determine liquid core length, initial droplet size, and velocity distributions, and how the presence of a liquid phase influence evaporation, mixing, and combustion?

Are there still droplets present, although the environmental conditions with respect to liquid oxygen are supercritical, and surface tension vanishes?

What is the effect of fluid property variations in near-critical regions on propellant atomization and mixing, and how the oxygen jet look like in these conditions?

Does the assumption of thermodynamic equilibrium still hold for evaporation modeling?

What are techniques of measuring reacting flows to be developed or adapted to these conditions?

Has thermal non-equilibrium to be taken into account when predicting Coherent Anti-Stokes Raman Scattering (CARS) temperatures?

To answer these questions, three cryogenic test facilities have been developed, Mascotte at ONERA in the Palaiseau center, M3 and P8 at DLR in the Lampoldshausen center.

4.3. The high-pressure cryogenic test facilities

4.3.1. The M3 test facility

For cryogenic hydrogen/oxygen atomization and combustion research, The M3 Micro combustor university type facility is optimum. Its prime features are a liquid nitrogen bath with propellant and injector temperatures of about 80 K, high-speed valves with operation times less than 5 ms to minimize startup and shutdown transients, and full optical access to the entire combustion chamber.

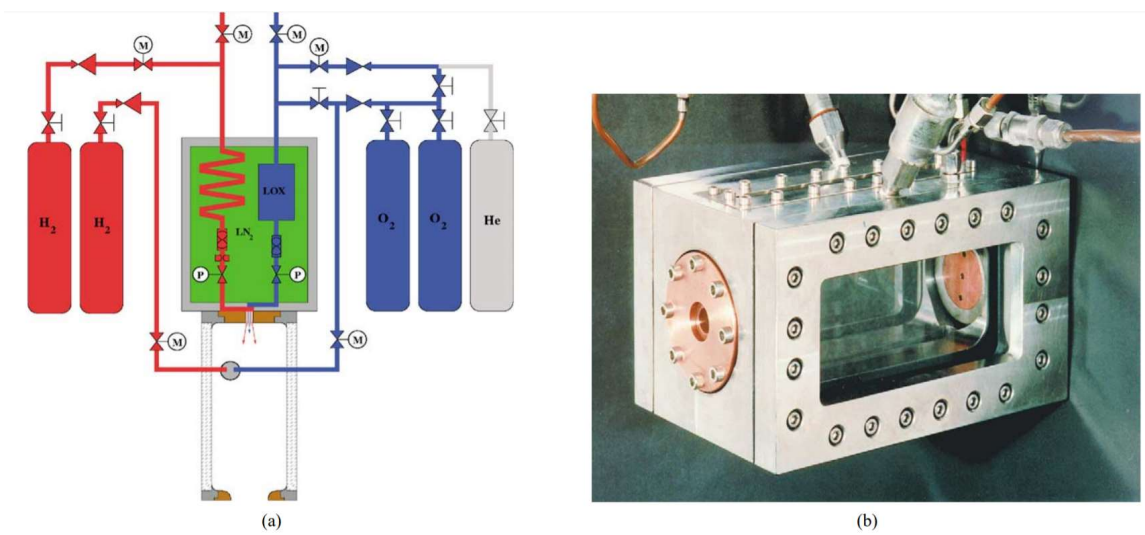


Figure 2 The M3 micro combustor facility. (a) Sketch of M3 flow schematic. (b) View of the rectangular combustion chamber.

4.3.2. The Mascotte test facility

Mascotte is an intermediate-range test bench between university laboratories, M3, and the P8 facility. Its development was achieved progressively, with increasing complexity. Three successive versions have been built:

1. only low chamber pressures (1 MPa) and hydrogen at room temperature were available

2. A heat exchanger was implemented in the hydrogen line in order to cool hydrogen down to -173 Degree, which is typically the injection temperature for the Vulcain engine thrust chamber. This enables the increase of the maximum flow rate of gaseous hydrogen (increased density) at low pressure without choking the injector exit

3. High pressures (10 MPa) in the combustor and at increasing the maximum LOX flow rate. It is the most important version, for a research test bench, with regard to the complexity of the bench components, the combustor technology, and the computing and monitoring systems—Mascotte operating range. The maximum chamber pressure with versions V01 and V02 were 1 MPa. V03 will permit operation at 10 MPa in the combustion chamber. Presently, the limiting factors are the technical issues associated with the design of a high-pressure combustor.

4.3.3. The P8 test facility

The P8 is the French/German facility for high-pressure hydrogen/oxygen combustion research and technology, in a joint effort of agencies, research laboratories, and industries both from France and Germany, operated by DLR. The facility has been equipped with additional systems such as high-pressure nitrogen and helium purge lines, a high-pressure hydrogen cooling line, and only recently with a high-pressure liquid hydrogen system. The facility is capable of operating under high variation in propellant and coolant mass flow rates and pressures at the interface. High-pressure mass flow regulation valves, equipped with roller spindles and actuators, facilitate adjustment of mass flows with a precision of 1% and also allow for quick changes in mass flows. In this way, different operating points can be adjusted during one test, and thus the overall number of tests can be reduced. Features such as test duration of up to 15 seconds at maximum mass flow rates, a possibility for cooling the hydrogen to the temperature of liquid nitrogen using a GH2/LN2 cooler, active hydrogen temperature control through the mixing of pre-cooled and ambient GH2. Model combustor C with optical access has a modular design that allows changing the axial position of the windows. Thus flow phenomena can be observed from the injector plane to 0.43 m downstream. This combustor is designed for chamber pressures up to 10 MPa, see Fig. 3.

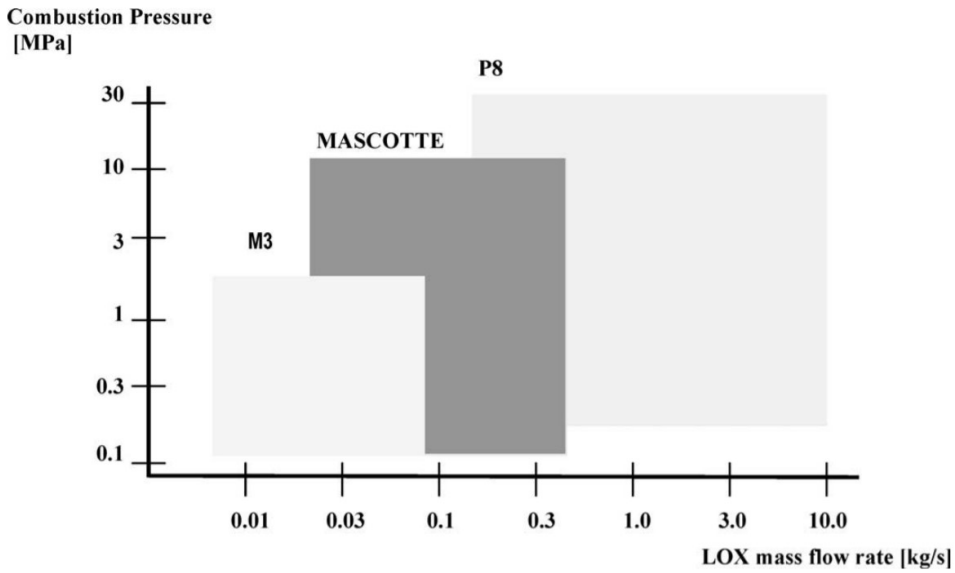


Figure 3. Pressure and mass flow rate domains of DLR and ONERA cryogenic combustion facilities

4.3.4. Other European test facilities

A major aim of the French/German MoU was to identify common scientific and technological problems and to define a joint complementary research strategy in order to enlarge the scientific outcome by taking advantage of synergetic effects. The increasing complexity of laser-optical diagnostic techniques and the high operating cost of facilities such as P8 makes it almost mandatory to develop such techniques and check their applicability and performance at both less demanding and less costly conditions. Hence, the complementary design in terms of pressure and mass flow rate levels of the cryogenic combustion facilities at DLR Lampoldshausen and ONERA Palaiseau, as shown in Fig. 5, is an outcome of the European spirit behind this MoU.

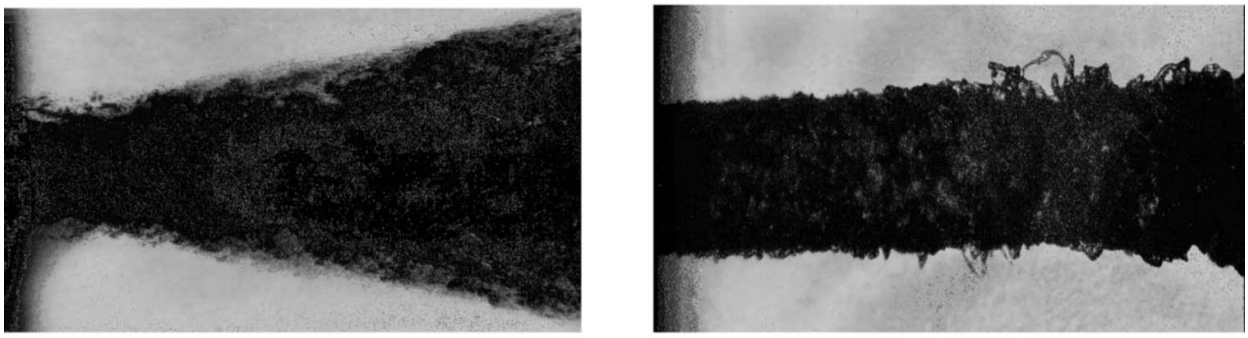
Section 5. High-pressure cryogenic combustion – examples of experimental achievements

Various test campaigns have been performed using the ONERA and DLR described above. This section provides some examples of experimental achievements on LOX/Hydrogen combustion.

5.1. Cold flow injection and atomization

For the determination of differences in the atomization pattern between cold and hot flow conditions, a couple of preliminary tests were performed at the M3 Micro Combustor, applying a standard shadowgraph setup. The atomization patterns of a liquid oxygen jet from a shear coaxial injector fed with liquid oxygen and

gaseous hydrogen at 80 K into the chamber at 0.14 MPa are shown in Fig. 4(a) cold case as in 4(b) for the burning issue. The initial diameter of the liquid oxygen jet at the combustor entrance is 1.2 mm, a velocity ratio of hydrogen/oxygen of about 17, and a momentum flux ratio of 0.1



(a) (b)

Figure 4 Shear coax atomization pattern; TGH2 = TLOX = 80 K, P = 0.14 MPa, DLOX = 1.2 mm, J = 0.1. (a) M3 Combustor atomization cold flow. (b) M3 combustor atomization with combustion

5.2. Spray combustion diagnostics

Experimental investigations have been performed to study injection and atomization under hot fire conditions. These diagnostic are essential because all structural properties and life predictions can be obtained. Jet breakup and atomization were studied qualitatively and quantitatively on Mascotte and M3, using high-speed visualization, Particle Imaging, and Velocimetry Phase Doppler Anemometry.

5.2.1. Phase Doppler Anemometry and Visualization

A test campaign was achieved under hot conditions, at 0.1 and 1 MPa in the Mascotte facility. First, to provide preliminary information on the fluid dynamics of the LOX core breaking process, a visualization technique using a stroboscopic laser sheet associated with a high-speed camera (2000 frames/s) was used. After these visualizations, droplet size and velocity measurements were performed by means of a Phase Doppler Particle Analyzer (PDPA) with a component of velocity.

The visualization phase was aimed at providing preliminary information on the LOX core breakup process and at localizing the atomization zone. Four operating conditions have been investigated during this test campaign. Two kinds of images were recorded during these visualization tests. First, phenomena were recorded without any filter to reject the luminous flame light. Those images permitted us to get the structures of both the luminous flame and the liquid inside the reaction zone. In a second configuration, a bandpass filter centered at the wavelength of the laser beam was fitted in front of the camera lens.

Another interesting piece of information was the comparison between results at 0.1 and 1 MPa. At 1 MPa, the flame emission was clearly detectable in the near blue spectral region while it was not at atmospheric pressure. This technique provides a qualitative idea of the secondary atomization process by considering a sequence of

images in the given zone of the jet. After the flow visualization tests, measurements of PDPA were performed. More than 100 tests were carried out at atmospheric pressure for the operating points point A and C. Seven radial profiles have been explored along the flow axis and as close as possible to the LOX post tip. Only two radial profiles have been obtained at 1 MPa, one very close to the coaxial injector, with x/d = 4 and another at x/d = 16. Here, like at atmospheric pressure, a measurement outside of the combustion chamber, after the throat, showed that no droplet existed at a distance of x/d = 83.6. The PDPA measurements at atmospheric pressure, along with high-speed visualizations, allowed us to get a relatively detailed description of the spray pattern, as shown in Fig. 5.

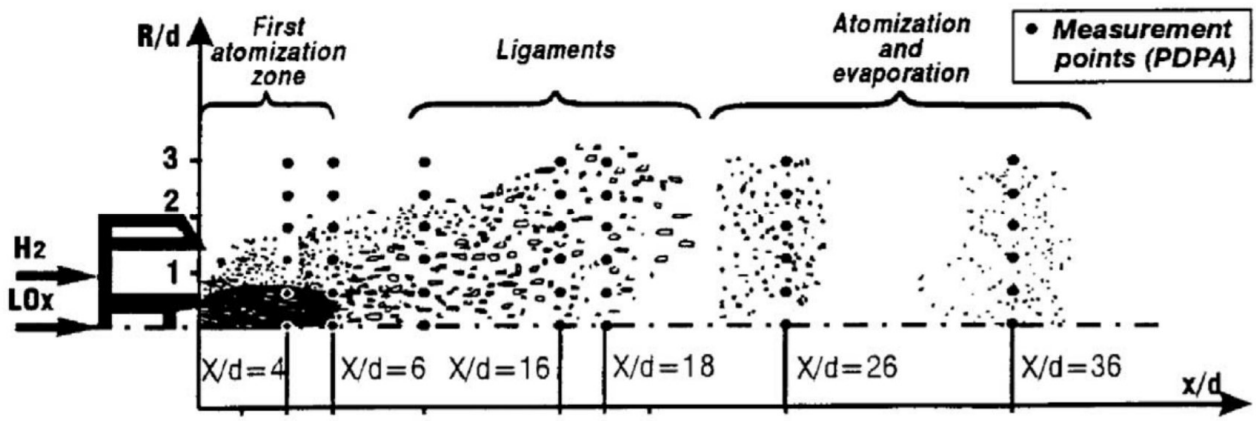


Figure 5 Qualitative description of the mean structure of the spray (0.1 MPa).

5.2.2. Particle Tracking Velocimetry

PTV is an optical method based on Particle Image Velocimetry (PIV) to measure the instantaneous velocity field using the atomizing liquid as a tracer. Although oxygen is only present in specific regions of the combustion chamber, the knowledge of both position and velocity of the liquid droplets gives a more detailed insight into the combustion chamber flow field. A comparison of the local velocities with the initial velocity of the liquid jet helps to qualify the aerodynamic forces on the liquid. Fig. 10 shows the instantaneous velocity field of a triple coaxial injector arrangement inside the M3 micro combustor with a momentum flux ratio J of 0.2 and a velocity ratio of hydrogen to oxygen 11.4.

The initial LOX diameter is 1 mm, with a total injector diameter of 3.6 mm. The distances between the injectors and between the outer injectors and the walls are equal. Fig. 6 combines eight single pictures from different experiments taken at different ent axial and radial positions. As known from flow visualization experiments, in the near injector region, the spreading angle of the spray is small. Hence, there are no tracer particles present for velocity determination. Furthermore, in the dense spray areas, the signal-to-noise ratio is too poor for a successful droplet validation algorithm. Nevertheless, For cryogenic spray combustion, a comparison of similar PTV results taken at different injection conditions or combustion chamber pressures gives some useful insight into the atomization and mixing mechanisms.

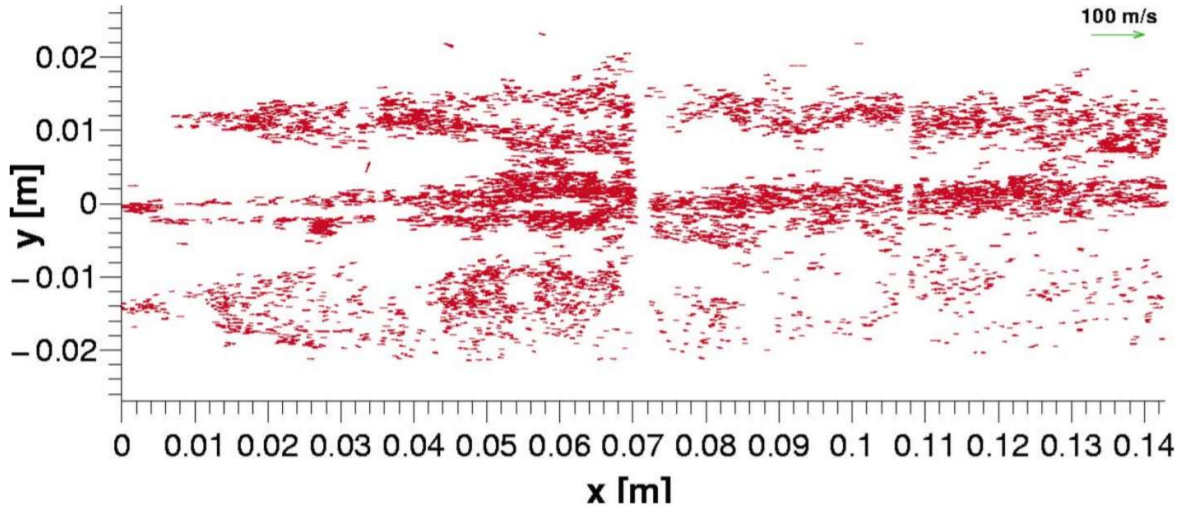


Figure 6 M3 micro combustor triple injector flow field at 0.5 MPa and J = 0.2

5.3. Combustion diagnostics for Turbulent systems

5.3.1. H2O emission spectroscopy

It is possible to use the spontaneous emission of the H2O to characterize the flame area as the water is a sub-product of the reaction. An example of a typical result for supercritical spray combustion is presented in Fig. 7. The operating conditions are as follows: combustion chamber pressure 6.0 MPa and injection velocities of oxygen and hydrogen of 20 m s−1 and 207 m s−1, respectively. These pictures have been recorded using an intensified CCD camera with an exposure time of 1 × 10−6 s. The following two instantaneous images show the near injector region with a time delay of 20 ms. It can clearly be seen that the product steam is already present directly at the exit of the injector, a finding which is in agreement with OH images taken during similar experiments.

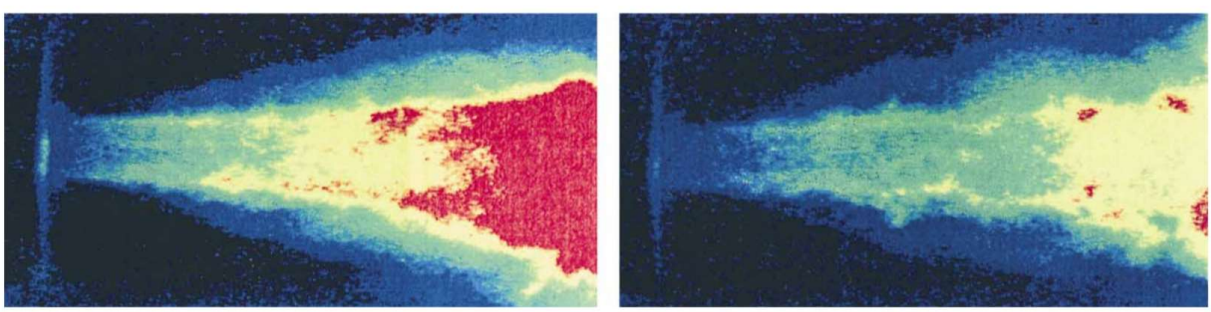


Figure 7. Typical H2O emission images of the near injector region in P8 model combustor C at 6 MPa.

5.3.2. OH radical imaging

The Mascotte facility and the GDR synthesis colloquium proceedings provided an extensive database of images under varied operating conditions. Some examples of

OH imaging are shown in Figs. 8. Emission of OH radicals is used to locate regions of intense combustion. Fig. 8 shows an example of OH emission images, and the injector geometry is also displayed. It is important to remember that the camera's signal is integrated over the line of sight, and the precise nature of the patterns appearing in these pictures can not be determined. It is, however, clear that radiation from OH starts right at the injector tips. The reactive layer is nearly cylindrical initially and expands further downstream.

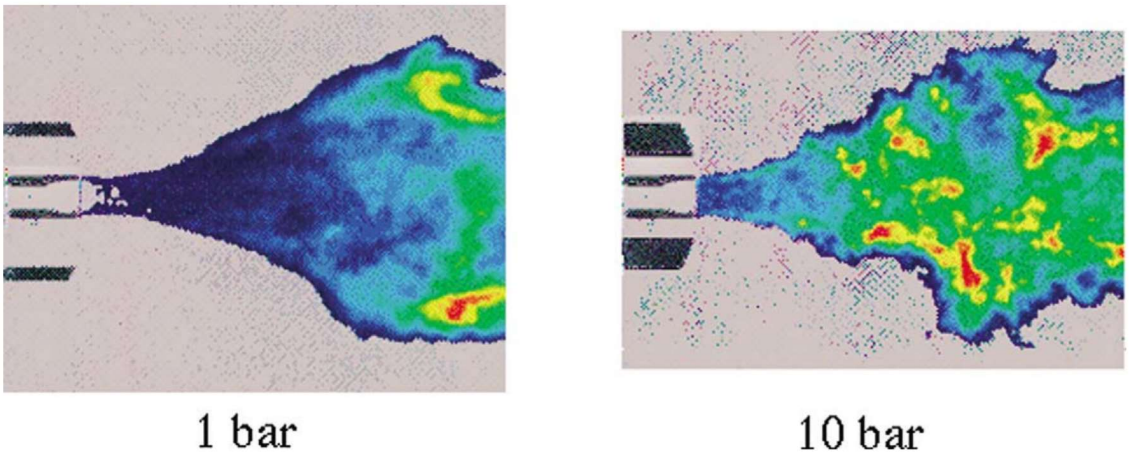


Figure 8. Typical instantaneous images of OH emission

In Fig. 9, the LIF signal is already quite strong at the injector, and the level does not change notably downstream. A highly corrugated layer of intense fluorescence can be visualized in the lower part of the picture while LIF's upper trace fades out. The signal does not entirely vanish in the LOX core, indicating that the camera also detects some parasitic motion. The presence or absence of LIF signals on the upper side of the chamber may be interpreted by considering the quality of LOX core disintegration and atomization. If this process is delayed, the laser sheet crosses the LOX jet without much distortion and produces OH fluorescence. This is the case in the injector vicinity. If the jet breaks up and creates a dense spray of droplets, the beam is scattered effectively and cannot produce fluorescence from the upper flame sheet. This explains why the LIF signal is much stronger on the lower side of the chamber, which is illuminated by the unperturbed laser sheet. This general picture is confirmed by the data deduced from the Abel transformed average emission images. Fig. 10 gives the averaged OH emission and the corresponding transformed image for operating condition A at 1 Mpa.

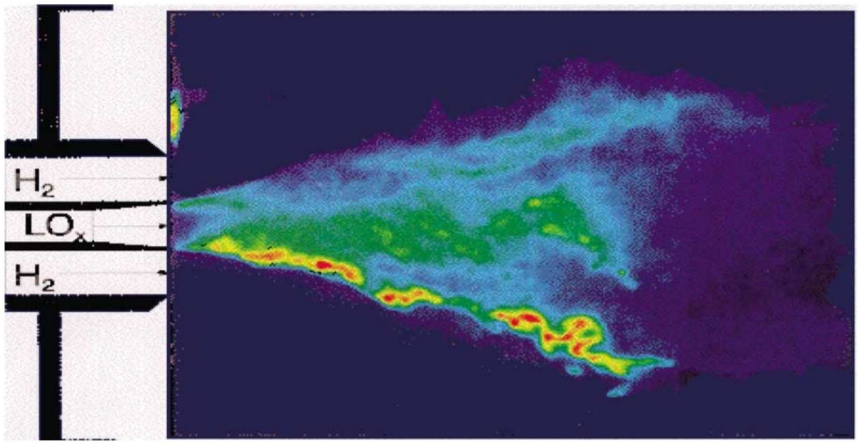


Figure 9. . Typical OH LIF image (the laser sheet is transmitted into the combustor from below).

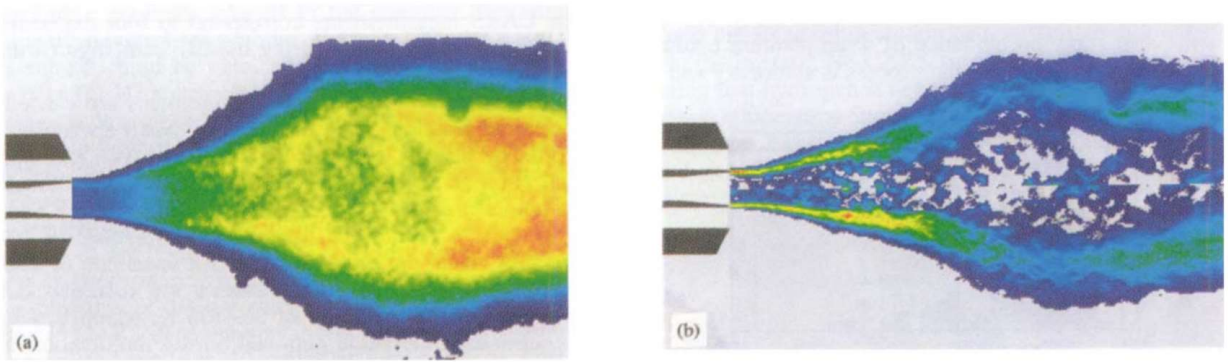


Figure 10 Average emission image (a) and Abel transformed emission image (b) Ref [15]

Section 6. High-pressure cryogenic combustion – examples of numerical and Modeling results

From the various numerical results available today, only two examples are given here in some detail. While the first example shows the simulation of a 0.5 MPa spray combustion at M3, the second gives the results obtained for the 1.0 MPa Mascotte spray combustion case. Prior to a description of these examples, the main features of the different CFD tools are as follows.

6.1. M3 Cryogenic spray combustion simulations

A low concentration spray is considered that is injected into a turbulent gaseous hydrogen stream where the inlet temperatures are cryogenic. The model includes an Eulerian description of the gas phase and Lagrangian equations for the dilute spray. The k−ε turbulence model is employed where additional terms account for the spray interaction. Chemical reactions are described through a flamelet model for turbulent spray diffusion flames. The conservation equation for the mixture fraction and its variance also accounts for mass gain through Vaporization; for details, see [16,17]. Convective heating and Vaporization are described through a model by Abramson and Sirignano [18]. The equation for droplet motion accounts for turbulent fluctuations; the spray distribution is described through the discrete droplet model, see [19,20].

Gas-phase characteristics in the cryogenic temperature regime are taken from JSME tables [21]. Furthermore, the pressure and temperature dependence of the vaporization rate and of the binary equilibrium composition at gas/liquid interface is included [22]. An example of the computational results achieved so far is given in Fig. 11(a), (b), which shows some characteristics of the liquid phase for a combustion chamber pressure of 0.5 MPa and a position of 104 mm downstream of the injector faceplate. Fig. 11(a), (b) show radial profiles of measured and predicted axial droplet velocities, respectively. The comparison shows that the agreement for both droplet velocities and radii is quite excellent.

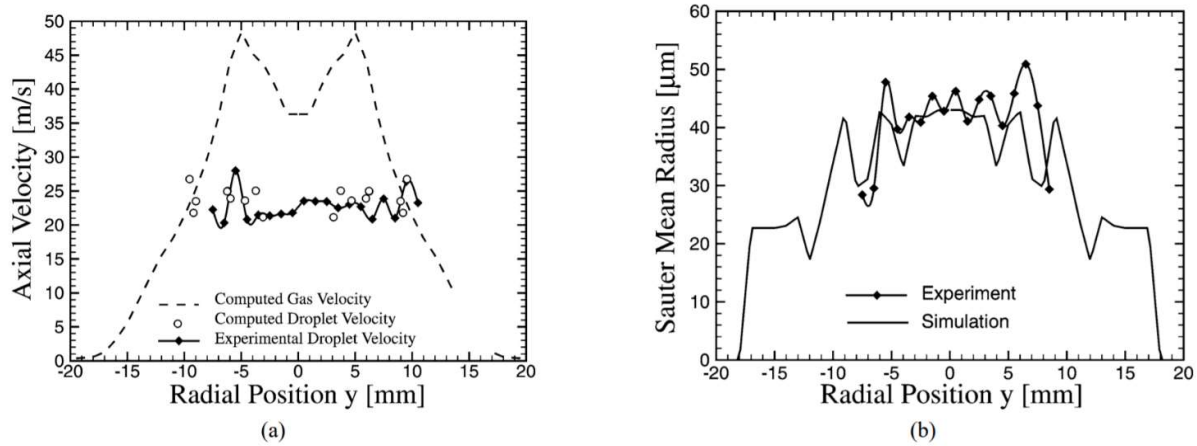


Figure 11. . Experimental and computational results from M3: P = 0.5 MPa, downstream position of x = 104 mm. (a) Comparison of axial droplet velocities

6.2. MASCOTTE Spray Combustion Simulations

The Flow field analysis of 1.0 MPa Mascotte was performed using the ONERA MSD code dedicated to propulsion systems. The unsteady, three-dimensional, Navier–Stokes, Reynolds-Averaged equations for a mixture of perfect gases are solved by MSD codes. The setup is based on finite volume techniques on structured curvilinear grids. The time integration can be either explicit; then a predictor-corrector scheme is used, or implicit with first or second-order accuracy. The implicit algorithm uses a classical ADI (Alternate Direction Implicit) factorization. The spatial discretization scheme is second-order accurate. The Euler fluxes are evaluated through a "Flux Difference Splitting" TVD scheme. For spray analysis, two solvers are available in the code: an Eulerian solver and a Lagrangian solver. In the present case, simulations were performed using the Lagrangian solver. Droplets are tracked in the turbulent flow, and a two-way coupling between the liquid phase and the gas phase is performed. Only one group of droplets was considered in the computations. Gas-phase turbulence is computed with the k−ε turbulence model. The turbulent dispersion is treated by the Gossman and Ionnides Eddy Life Time dispersion model with an additional spatial decorrelation criterion [3] to better account for the crossing trajectory effect.

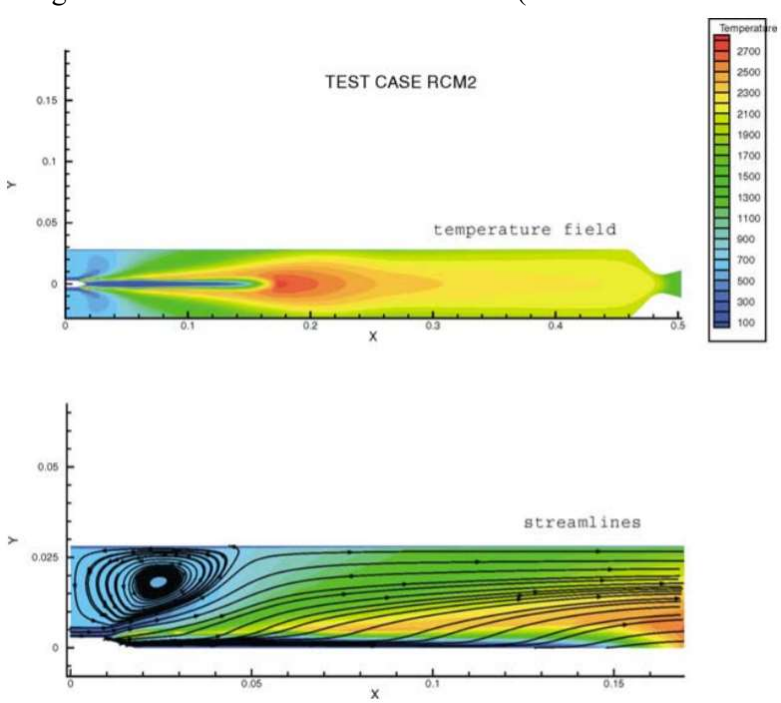


Figure 12. Temperature field and streamlines – CLE model. Droplet size

Vaporization is computed with the standard "D2" model with Ranz–Marshall correction to account for convection around the droplet. Combustion was described using the CLE model, which is a turbulent combustion model assuming an infinitely fast single scalar chemistry with a β-function pdf and a thermodynamical equilibrium limitation. Fig. 12 shows the temperature field and streamlines, where the recirculation zone can be observed. The agreement between experiments and computations is very good.

Section 7. Analysis and Solutions

7.1 Thermomechanical Analysis and optimization

The cryogenic hydrogen/oxygen-powered Vulcain engine is the main stage engine of the European Ariane 5 space transportation system. The regeneratively cooled gas generator cycle engine structure can be seen in Figure.13. The peak heat load in the throat area exceeds 80 MW/m2 at the nominal operating point of 11 MPa. The regenerative cooling introduces an enormous temperature gradient across the combustion chamber walls yielding large heat flux and thermal strains. The

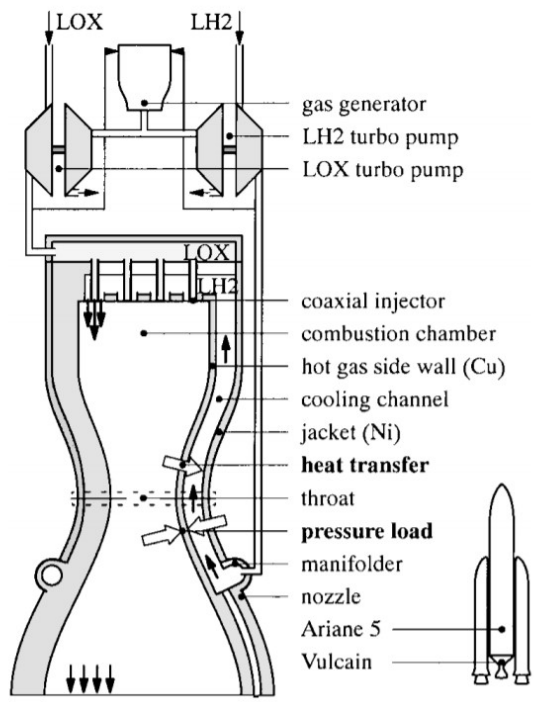


Figure 13 Schematic view of the regenerative cooled combustion chamber vulacin

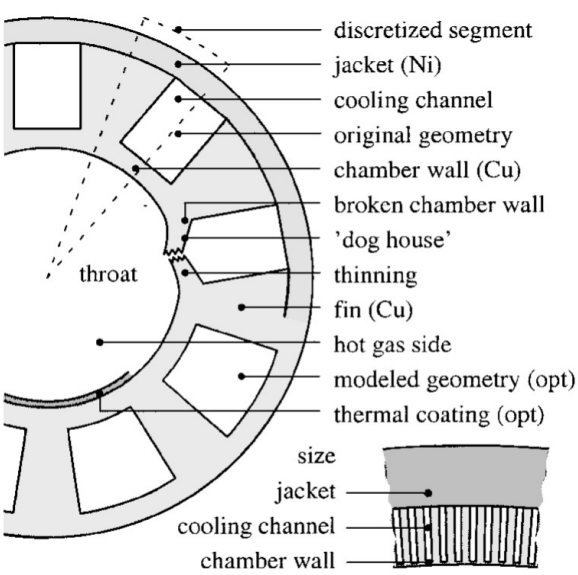


Figure 14 Throat of the combustion chamber: modeling, characteristic failure, and optimization model

resulting plastic deformations of the wall material limit the combustion chamber lifetime. Experimental investigations by Quentmeyer[10,11]have demonstrated that low-cycle fatigue and creep are the main reasons for material failure. Figure.14. shows a combustion chamber cross-section emphasizing characteristic features of a regeneratively cooled combustion chamber such as chamber wall, failure location, cooling channel geometry, and outer liner materials.

7.1.1. Analysis Motive

The increasing requirements for the performance of cryogenic liquid rocket engines in combination with the need to

reduce the operational cost continuously have led to the idea of reusable space transportation systems. Because all reusable systems are based on an increased combustion chamber lifetime, the current research and technological activities at DLR, Lampold shausen, aim at the following

1) Detailed understanding of failure mechanisms.
2) Development of a tool to predict the life of the combustion chamber.
3) An optimized design of the combustion chamber.

Before these items can be realized, the general problem of heat and mass transfer in a combustion chamber of a liquid rocket engine needs to be solved: the prediction of injection, atomization, mixing, and combustion of the propellant, as well as the hot-gas flow in the combustion chamber, the heat transfer from the hot-gas side to the wall, further into the structure, into the coolant, finally, the coolant flow itself. The basis of a deeper understanding of the failure mechanisms of current combustion chamber designs is detailed experimental data, material, and failure laws, as well as numerical simulations of the structural behavior[12,13]. The inverse thermomechanical problem is used for automatic design optimization. Therefore, optimization methods will be applied to improve future developments of combustion chambers[14]. The combination of the design tools and results yields an improvement of both coolant-side heat transfer and the combustion chamber geometry and increases the lifetime of cryogenic liquid rocket engines.

7.2.2. Direct Thermomechanical Problems

The simulation and Modeling of the direct thermo-mechanical problem are focused basically on the coolant flow and deformation of the combustion chamber wall. The introduced effect of the combustion chamber flow and heating on the hot-gas side is due to prescribed boundary conditions, named as the combustion chamber pressure and the hot-gas-side heat flux.

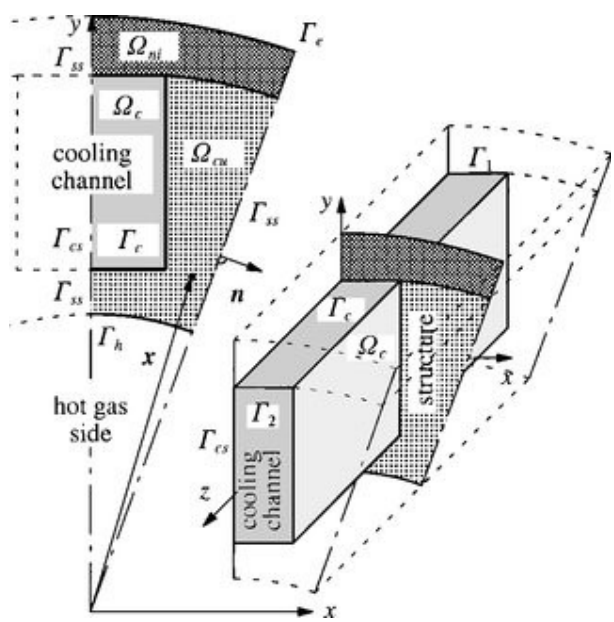


Figure 15. Computational model of the combustion chamber.

The thermomechanical fluid-structure model shown is sub-divided into the coupled fluid flow and thermal field problem and the thermomechanical structural problem, whereby the parts are connected by the heat transfer and the pressure pc at the shared boundary[15]. The cooling channel and the combustion chamber wall geometry is changed by a representative geometry to make the model simple, given by a three-

dimensional straight cooling channel and thermal, structural model and plane stress thermomechanical structural model as shown in Fig.3.

7.2.3. Optimization

The inverse problem can be solved by the direct problem described earlier. The framework of the material, loads, and inverse problem, as well as manufacturing restrictions, are given. A way to obtain an optimal behavior is obtained for the design of the combustion chamber with respect to a well-defined objective. The methods of mathematical programming help in solving inverse problems described earlier, where the solution of the direct problem is included in determining the objective and its sensitivity with respect to the design parameters. Practically, the application of optimization methods is restricted to an additional optimization loop in the program code, including, as the core, the solution procedure of the direct problem. Consequently, these methods may be applied simply to calculate the optimal design of combustion chambers. Within this paper, the used optimization method and the application to the combustion chamber design are investigated by means of a simplified linear thermomechanical model of the combustion chamber.

7.2 Numerical Simulation of LRE

7.2.1 Heat Analysis

In a liquid-propellant rocket engine, cryogenic/thermal insulation is designed to protect and prevent the engine components. Cryogenic insulation is mainly required to protect from the heat and maintain the temperature, -183℃(90K) to avoid Vaporization. The simple numerical simulation is performed using the governing equation as following

This simple 1-D analysis is applied to each component and decides the material and thickness of cryogenic/thermal insulations.

LRE	Cryogenic Insulations
VULCAIN	LOX/LH2 pipeline: polyurethane foam + AL Tape
LE-7A	LOX/LH2 pipeline, TP: polyurethane foam + AL Tape
HM-7B	LOX/LH2 pipeline: polyurethane foam + AL Tape
J2-X	LOX/LH2 pipeline: polyurethane foam + AL Tape
MERLIN	LOX pipeline: polyurethane foam + AL Tape
KSR-III	polyurethane

Table 1 Cryogenic Insulation types in LRE

The 1-D governing equation on heat transfer is simulated for insulation analysis. In equation (1), no heat generation and steady-state are canceled out terms. From (1), equation (2) is integrated, and qr is generated from the combination of (2) and (4) with the assumption Ti ≈ T1. This simple 1-D analysis is applied to each component and decides the material and thickness of cryogenic/thermal insulations.

7.2.2 Material Selection

As the numerical analysis determines insulation thickness and temperature distribution, several candidate's cryogenic insulations are tested through liquid nitrogen experiment comparing with the numerical analysis. Liquid nitrogen is substituted due to the reason relatively safe and easy-handling comparing to liquid oxygen. The ideal cryogenic insulation material is Aerogel (cryogel z) to protect from heat flux and maintain liquid state temperature for oxygen by the numerical analysis. However, Aerogel (cryogel z) contains nano-sized particles, which is not suitable for LRE and has no flexibility to apply on the engine components' surface. Many other cryogenic insulations are used in the international rocket engine industry and applied to protect and maintain the cryogenic temperature. For heat radiation from the exhaust plume, a nickel-foil/refrasil blanket was insulated. The polyurethane foam was mostly applied to the liquid oxygen pipeline after all engine assembly was completed in table 1. The polyurethane foam has disadvantages that are no reusable, dust occurrence, hard managing insulation thickness.

The several cryogenic insulations with similar properties to Aerogel (cryogel z) are tested in a liquid nitrogen test (figure 15). The inner and outer temperatures of each cryogenic insulation with time are recorded and compared. The liquid

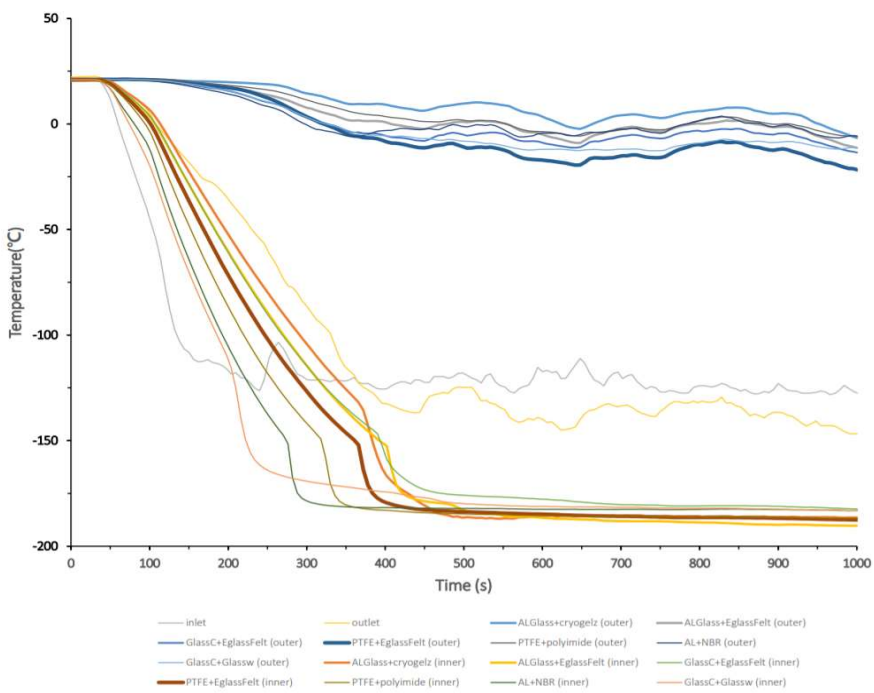


Figure 16. Cryogenic Insulation Test using liquid nitrogen

nitrogen test materials are glass felt, NBR, glass wool, and cryogel z with cover material PTFE, ALcoating glass, and glass coating. The cryogenic insulation, glass felt type with ALcoating glass cover is selected best close to Aerogel (cryogel z). ALcoating glass cover helps to remain the cryogenic temperature and solving the icing problem. Also, it protects the radiation from the heat up to 400 ℃ (glass material) LRE (Liquid-propellant Rocket Engine) are commonly clustered into more than 3-4 engines to create combined thrust in 1 stage.

When LRE clustering, the engine exhaust plume by several engine exhaust flames is created. A large amount of heat and radiation affects the engine itself, especially in the middle of engine clustering. The heat shield exists for this problem, but each MFV of KSLV-II LRE is exposed to the exhaust plume below the heat shield, although the fuel in MFV keeps running and flowing into the combustor. In high-temperature thermal insulation, the insulation thickness and temperature distribution from the numerical Analysis considering Turbopump manifold temperature by actual engine test determines the material and installation method. Figure 16 shows the comparison between silica type thermal insulation and the required insulation specification. Silica type and alumina-silica type (up to 1600 ℃) are selected for turbopump manifold and MFV protection from exhaust plume while clustering.

7.3 Thrust Chamber Analysis

Thrust chamber of cryogenic engine is modeled at a thrust of 50KN and chamber pressure of 40 bar to reduce the high temperature and pressure in the combustion chamber. The performance parameters are obtained through NASA-CEA code for the given input condition of the engine. To verify the design quality of the field investigation of the nozzle, nozzle flow has to be studied. A 2-D axis-symmetric computational simulation of the flow field is required, either developing a code or using commercial code. In the current study, computation has been made to obtain flow through the nozzle using commercial software FLUENT. In the present study, two-dimensional structured grids were generated on the nozzle profile and an extended domain. For axis-symmetric computation, quadrilateral cells with map scheme were used for grid generation

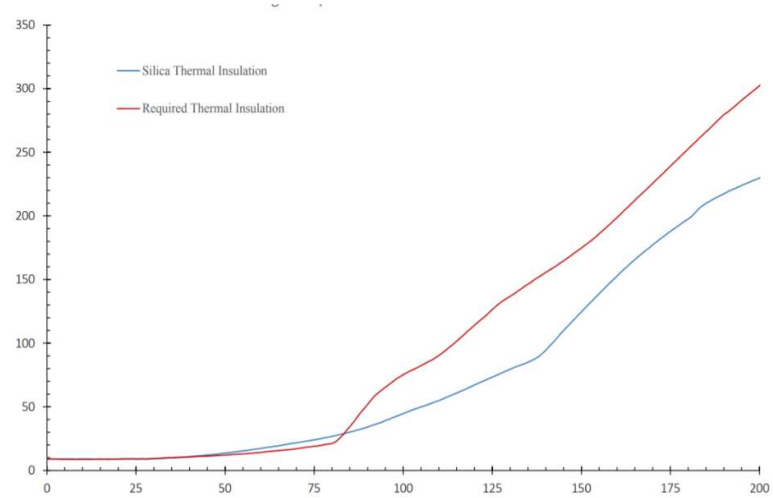


Figure 17. Comparison between Silica-type Thermal Insulation and Required Insulation Specification

7.3.1 Boundary Conditions

The boundary conditions values are taken from the NASA-CEA code.

Pressure inlet Total pressure = 40 bar

Total temperature = 3420 Kelvin

Specific heat = 3768 J/Kg-K

Viscosity = 1.0455E-4 Kg/m-s

Thermal conductivity = 0.5590 W/mK

Molecular weight = 13.37

7.3.2 Analysis Results of Nozzle flow

The following results contain the information about the nozzle pressure, velocity, temperature, and mach no variation along with the nozzle and domain for LOX/LH2 propellant combination.

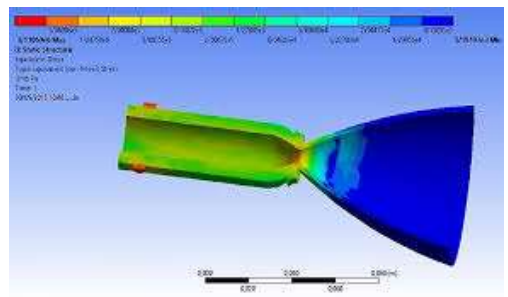


Figure 18 Structural Analysis

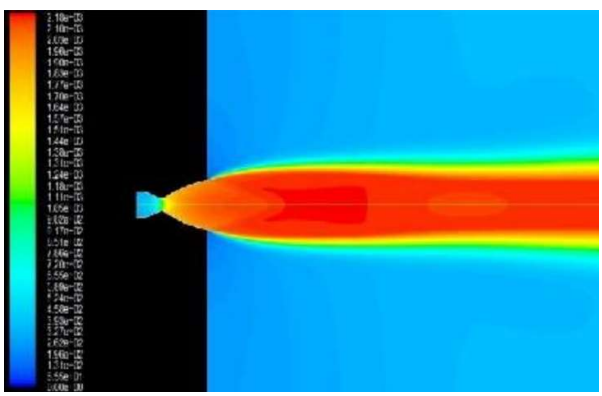


Figure 21. Contour of Velocity Magnitude (m/s)

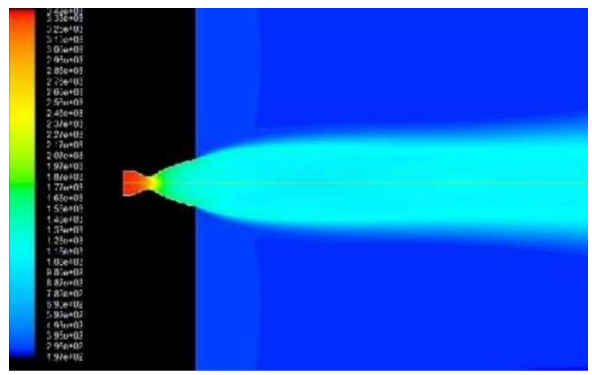


Figure 20. Contour of Static Temperature (K)

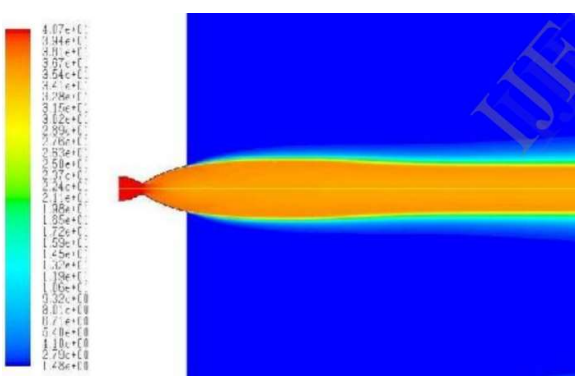


Figure 19. Contour of Total Pressure

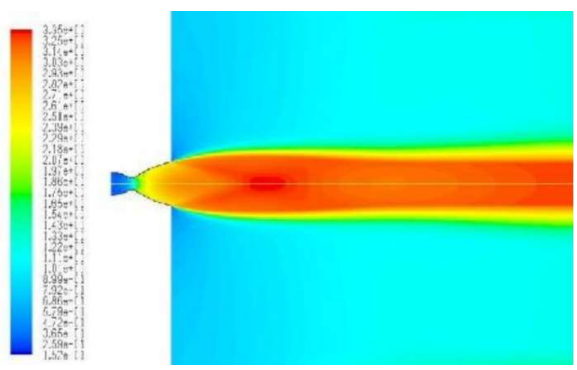


Figure 22.. Contours of Mach Number

Section 7. Conclusion and outlook

European research activities on high-pressure cryogenic combustion have been discussed. They aim to get insight into complex processes involved in liquid rocket combustion, improving modeling and simulation tools. The main objective is to provide European industry with efficient tools to design high performance and reliable engines. Strong cooperation has been set between France and Germany on cryogenic combustion. We know not even better the advantages and shortcomings of state-of-the-art CFD tools when applied to problems of cryogenic liquid rocket engines; we have even been able to improve those tools incorporating experimental results obtained in the frame of the cooperation.

- The experimental data indicate clearly that typical H2/LOX flames are directly anchored at the tip of the LOX post in contrast to what is generally observed for hydrocarbon flames, which are lifted-off. This finding has been verified for a hydrogen temperature ranging from ambient down to 80 K.
- The flame's general structure can now be characterized as a thin sheath around the liquid or dense gaseous oxygen where hardly ever liquid ligaments or droplets cross this boundary.
- Liquid core's length is obtained for a large variety of operating conditions, and, based on this observation, an empirical correlation has been derived. Although it is known that the dense core length decreases with increasing pressure, a detailed description is still missing.
- Heat transfer to fluids near their critical pressure yields an increase in a specific volume and not in temperature, a phenomenon which may be described as "quasi-boiling." Nevertheless, the experimental results obtained so far do not yet allow a conclusion about the influence of this iso-thermal expansion on local transport and mixing processes. Also, in the advancement of the theoretical models for interpretation of these highly nonlinear signals too. The database available now has led to considerable improvements in the Modeling of identified dominating physical phenomena as well as their incorporation in CFD-based design tools for cryogenic liquid rocket engines. Despite efforts spent so far, no satisfactory atomization model is available to describe in detail shear-coaxial jet breakup. This topic is even more critical because droplet data (drop size and velocity distributions) are boundary conditions for spray numerical simulations. At high pressures, i.e., above the critical oxygen pressure, more data are still needed to characterize cryogenic flames and to calibrate CFD tools.
- Finally, in the framework of new European development programs for expendable and future reusable launchers, other issues arise, such as ignition, high-frequency combustion instability, and combustion of alternative propellants. The European cooperation will be extended and enhanced through these new topics of common interest.
- Within the framework of the presented method, the discretization and solution of the fluid and solid equations were realized by nonlinear finite element methods
- KSLV-II is in the middle of the developmental stage of Liquid-propellant Rocket Engine (LRE), so that the essential and decisive numbers and configurations are confidential and classified yet. The temperature distribution on components is analyzed, and the materials and thickness of hot/cryogenic

insulation are selected. The optimization of insulations is required to make provision hot/cryogenic temperature. The application of insulations is essential to each component by materials, configurations, fasten, and connection to increase the insulated area's surface.

- Cryogenic propellants in liquid rocket engines provide a high specific impulse suitable for use in rocket booster stages. The thrust chamber modeled at 50KN of thrust and chamber pressure of 40 bar with the propellant combination of LH2/LOX shows a very great consistency when compared with the computational results.

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